

# GTIIT Spring 2025 Calculus 1M

## Workshop 6

April 9, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

**If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!**

**Important note:** If you find any errors or typos, please let me know so I can correct them.

1. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable functions, and let  $F$  be a function such that  $F'(x) = f(x)g'(x)$  and  $G$  be a function such that  $G'(x) = g(x)f'(x)$ . Show that:

$$F(x) = f(x)g(x) - G(x) + C$$

for some constant  $C$ .

2. If  $a$  and  $b$  are positive numbers, find the maximum value of  $f(x) = x^a(1-x)^b$ ,  $0 \leq x \leq 1$ .
3. Find the critical points of the functions:
  - (a)  $f(x) = x^2e^{-3x}$ ;
  - (b)  $f(\theta) = 2\cos(\theta) + \sin^2(\theta)$ ;
4. Show that the equation  $x^3 - 15x + c = 0$  has at most one root in the interval  $[-2, 2]$ .
5. Show that the equation  $x^4 + 4x + x = 0$  has at most two real roots.
6.
  - (i) Show that a polynomial of degree 3 has at most three real roots.
  - (ii) Show that a polynomial of degree  $n$  has at most  $n$  real roots.
7.
  - (a) Find the intervals on which  $f$  is increasing and decreasing;
  - (b) find the local maximum and minimum values of  $f$ ;
  - (c) find the intervals of concavity and the inflection points of  $f$ ;
  - (d) use the above information to sketch the graph of  $f$ .
  - $f(x) = x\sqrt{6-x}$ ;
  - $f(x) = x^{1/3}(x+4)$ .